

Written report

Group [REDACTED]

Task1 - Data Cleaning

We deleted useless content, such as Chinese characters and content irrelevant to our analysis, and retained district, longitude and latitude information, area, address, etc. After data cleaning, it is easier for us to analyze the data.

Task 2 – Database Design and Implementation

We have ten tables: Badminton Courts, Basketball Courts, census, Country Parks, Fitness Rooms, Park Zoos and Gardens, Sports Ground, Swimming Pools, Tennis Court, hk_admin. We create the table by using these SQL queries:

```
drop table if exists census;
create table census (
  district varchar(30),
  t_pop int,
  pop_m int,
  pop_f int,
  age_1 int, -- <15
  age_2 int, -- 15 - 24
  age_3 int, -- 25 - 44
  age_4 int, -- 45 - 64
  age_5 int, -- 65+
  -- M.A. stands for median age
  t_ma decimal(3, 1),
  ma_m decimal(3, 1),
  ma_f decimal(3, 1),
  primary key(district)
);

drop table if exists badminton_courts;
create table badminton_courts (
  address varchar(256),
  fac_type varchar(30),
  lat decimal(11, 8),
  lon decimal(11, 8),
  fac_name varchar(256),
  num int,
  district varchar(30),
  facilities varchar(256)
);

drop table if exists basketball_courts;
create table basketball_courts (
  address varchar(256),
  fac_type varchar(30),
  lat decimal(11, 8),
  lon decimal(11, 8),
  fac_name varchar(256),
  num int,
  district varchar(30),
  facilities varchar(256)
);

drop table if exists country_parks;
create table country_parks (
  fac_name varchar(50),
  easting decimal(10, 4),
  northing decimal(10, 4)
);

drop table if exists park_zoos_and_gardens;
create table park_zoos_and_gardens (
  address varchar(256),
  fac_type varchar(256),
  lat decimal(11, 8),
  lon decimal(11, 8),
  fac_name varchar(256),
  fac_desc varchar(384),
  district varchar(30),
  facilities varchar(256)
);

drop table if exists fitness_rooms;
create table fitness_rooms (
  address varchar(256),
  fac_type varchar(30),
  lat decimal(11, 8),
  lon decimal(11, 8),
  fac_name varchar(256),
  district varchar(30),
  facilities varchar(256)
);

drop table if exists sports_grounds;
create table sports_grounds (
  address varchar(256),
  fac_type varchar(30),
  lat decimal(10, 4),
  lon decimal(11, 8),
  fac_name varchar(256),
  district varchar(30),
  facilities varchar(256),
  area decimal(10, 4)
);

drop table if exists swimming_pools;
create table swimming_pools (
  address varchar(256),
  fac_type varchar(30),
  lat decimal(11, 8),
  lon decimal(11, 8),
  fac_name varchar(256),
  district varchar(30),
  facilities varchar(256)
);
```

```

drop table if exists tennis_courts;
create table tennis_courts (
    address varchar(256),
    fac_type varchar(30),
    lat decimal(11, 8),
    lon decimal(11, 8),
    fac_name varchar(256),
    fac_desc varchar(256),
    num int,
    district varchar(30),
    facilities varchar(256)
);

copy census from 'C:\Users\Public\Documents\Postgres16_Group\census.csv'
delimiter ',' csv header;

copy badminton_courts from 'C:\Users\Public\Documents\Postgres16_Group\BadmintonCourts.csv'
delimiter ',' csv header;

copy basketball_courts from 'C:\Users\Public\Documents\Postgres16_Group\BasketballCourts.csv'
delimiter ',' csv header;

copy country_parks from 'C:\Users\Public\Documents\Postgres16_Group\CountryParks.csv'
delimiter ',' csv header;

copy fitness_rooms from 'C:\Users\Public\Documents\Postgres16_Group\FitnessRooms.csv'
delimiter ',' csv header;

copy park_zoos_and_gardens from 'C:\Users\Public\Documents\Postgres16_Group\ParkZoosandGardens.csv'
delimiter ',' csv header;

copy sports_grounds from 'C:\Users\Public\Documents\Postgres16_Group\SportsGrounds.csv'
delimiter ',' csv header;

copy swimming_pools from 'C:\Users\Public\Documents\Postgres16_Group\SwimmingPools.csv'
delimiter ',' csv header;

copy tennis_courts from 'C:\Users\Public\Documents\Postgres16_Group\TennisCourts.csv'
delimiter ',' csv header;

-- Constructs an HK1980 point, converts to Wgs84
Drop Function If Exists point_hk1980(Numeric, Numeric);
Create Function point_hk1980(easting Numeric, northing Numeric)
Returns Geometry As $$
Begin
    Return ST_Transform(ST_SetSRID(ST_MakePoint(easting, northing), 2326), 4326);
End;
$$ Language plpgsql;

-- Constructs an Wgs84 point
Drop Function If Exists point(Numeric, Numeric);
Create Function point(lat Numeric, lon Numeric)
Returns Geometry As $$
Begin
    -- WGS84 uses (lon, lat) format
    Return ST_SetSRID(ST_MakePoint(lon, lat), 4326);
End;
$$ Language plpgsql;

-- Example
-- ...
-- With geom as (
--   Select geom From hk_admin Where district = 'NORTH'
-- )
-- Select name_en From basketball_courts, geom Where ST_Contains(geom, point(latitude, longitude))
-- ...

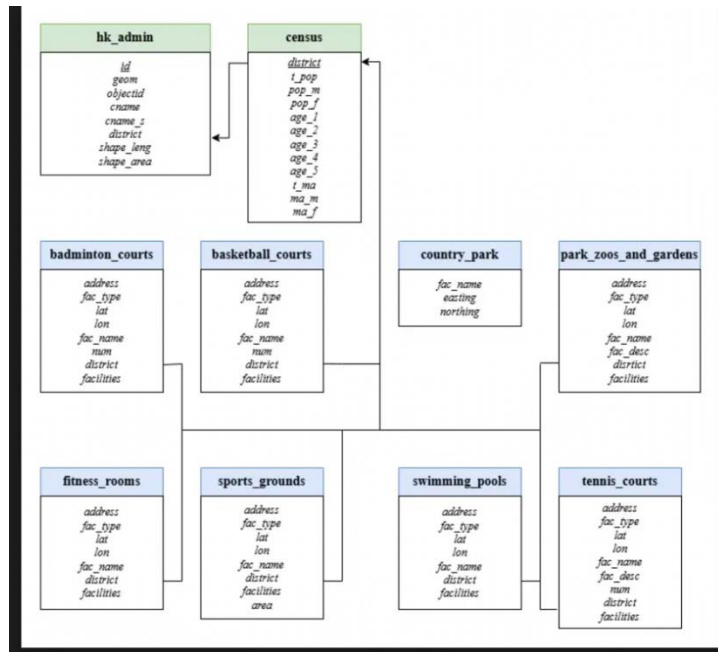
-- Converts meters to degrees
Drop Function If Exists meters(m Float);
Create Function meters(m Float)
Returns Float As $$
Begin
    Return m / 111319.9;
End;
$$ Language plpgsql;

```

We first create a table for each facility, then we import the table from the csv. The `drop`

statements are to ensure the current table is up to date due to possible changes of the SQL code. We also define some helper functions like `point` and `point\_hk1980` to create a geometry from WGS84 coordinates or HK1980 coordinates respectively. The `meters` function is to convert meters to longitudinal degrees.

The schema design of the relational database:



primary key:

District is the primary key for census, and other relations reference district.

Relation names:

Hk_admin: relation that stores the geometry of the administrative regions of Hong Kong

Census: the census data of Hong Kong

Facility relations: badminton_courts, basketball_courts, country_parks, park_zoo_and_gardens, fitness_rooms, sports_grounds, swimming_pools, tennis_courts

Each sports facility is a separate relation

Attributes:

In general

Address, fac_type (facility type), lat, lon (WGS84 coordinates), fac_name (facility name), district

Other

Facilities (facilities delimited by comma), num (number of facilities), fac_desc (facility description)

Task 3 – Perform Spatial and Non-Spatial Queries

– Given a geographic location (e.g., a lat/lng pair), find the nearest health facility and its relevant information (e.g., facility type, district, facilities)

```
with nearest as (select fac_type, lat,lon, district,facilities
From Badminton_Courts union
select fac_type, lat,lon, district,facilities
From swimming_pools union
select fac_type, lat,lon, district,facilities
From basketball_courts union
select fac_type, lat,lon, district,facilities
From tennis_courts union
select fac_type, lat,lon, district,facilities
From fitness_rooms union
select fac_type, lat,lon, district,facilities
From sports_grounds)
select fac_type, district,facilities
From nearest
order by ST_Distance(ST_SetSRID(ST_MakePoint(114, 22), 4326),
point(lat, lon))
limit 1
```

	fac_type character varying (30) 🔒	district character varying (30) 🔒	facilities character varying (256) 🔒
1	Recreation Grounds	ISLANDS	TENNIS COURTS

We set the location to (114, 22) and we find the nearest health facility is a tennis court on recreation grounds in the islands district.

Since we made each sports ground a separate relation, we use union on each health facility.

– Given a geographic location, how many types of sports and outdoor facilities are within a reasonable walking distance? (Students can assume a reasonable walking distance on their own).

```

with type_ as (select fac_type, lat,lon, district,facilities
From Badminton_Courts union
select fac_type, lat,lon, district,facilities
From swimming_pools union
select fac_type, lat,lon, district,facilities
From basketball_courts union
select fac_type, lat,lon, district,facilities
From tennis_courts union
select fac_type, lat,lon, district,facilities
From fitness_rooms union
select fac_type, lat,lon, district,facilities
From sports_grounds union
select 'Country Park' as fac_type,
ST_Y(ST_Transform(ST_SetSRID(
ST_MakePoint(easting, northing), 2326), 4326)) as lat,
ST_X(ST_Transform(ST_SetSRID(
ST_MakePoint(easting, northing), 2326), 4326)) as lon,
(select district
from hk_admin, country_parks
where ST_Contains(geom, point_hk1980(easting, northing))
limit 1) as district,
'' as facilities
From country_parks)
select count(distinct fac_type) as number_of_types from type_
where st_dwithin(st_setsrid(st_makepoint(114.181, 22.314), 4236),
st_setsrid(st_makepoint(lon, lat), 4236),
meters(1000)
);

```

	number_of_types bigint 
1	4

We set the location to (114.181, 22.314), and the walking distance to 1000m. We use union on all the relations, and one of the relations does not have the WGS84 coordinates, so we have to convert it into WGS84. Finally, we count the number of distinct facility types, where the geometry is within 1000 meters of the desired point.

– For a given type of sports or outdoor facilities, order the administrative districts in Hong Kong by the number of such facilities

	district character varying (30)	count_ bigint
1	KOWLOON CITY	4
2	SAI KUNG	4
3	WAN CHAI	4
4	ISLANDS	5
5	SOUTHERN	5
6	CENTRAL AND WESTERN	5
7	TSUEN WAN	6
8	TAI PO	6
9	EASTERN	6
10	SHAM SHUI PO	6
11	WONG TAI SIN	7
12	TUEN MUN	7
13	YAU TSIM MONG	7
14	SHA TIN	7
15	YUEN LONG	8
16	KWAI TSING	9
17	NORTH	10
18	KWUN TONG	10

```
select district, count (*) as count_
From Badminton_Courts
group by district
order by count_;
```

– For a given type of sports or outdoor facilities, order the administrative districts in Hong Kong by the per capita of facility count.

Badminton courts are used in this example. We select the district and the count from badminton courts, merge with census and divide the population by the number of badminton courts to get one badminton court per capita of facility count for each district in Hong Kong

```
with types_ as (select district, count(*) as count_
From Badminton_Courts b
group by district)
select distinct t.district, c.t_pop::float/count_ as per_capita
From types_ t, census c
where t.district=c.district
order by per_capita;
```

	district character varying (30) 🔒	per_capita double precision 🔒
1	NORTH	30963.1
2	ISLANDS	37056.4
3	WAN CHAI	41673.75
4	YAU TSIM MONG	44378.142857142855
5	CENTRAL AND WESTERN	47190.6
6	SOUTHERN	52655.6
7	TAI PO	52745
8	TSUEN WAN	53349
9	KWAI TSING	55088.666666666664
10	WONG TAI SIN	58114.57142857143
11	KWUN TONG	67316.6
12	SHAM SHUI PO	71848.33333333333
13	TUEN MUN	72411.28571428571
14	YUEN LONG	83510
15	EASTERN	88267.16666666667
16	SHA TIN	98972.28571428571
17	KOWLOON CITY	102658.5
18	SAI KUNG	122259.25

Addition questions

() The population density of people under the age of 15 using parks in each district of Hong Kong.

We can find how many children in each district use a garden or zoo on average. The reason for choosing this question is that among government sports facilities, the rate of use of gardens or zoos by small children is the highest.

```

1  SELECT p.district, (c.age_1/COUNT(*)) as a
2  FROM park_zoos_and_gardens p, census c
3  WHERE p.district = c.district
4  GROUP BY p.district, c.age_1
5  ORDER BY a;

```

Data Output Messages Notifications		
	district character varying (30)	a bigint
1	WAN CHAI	2114
2	CENTRAL AND WESTERN	2438
3	SOUTHERN	4542
4	ISLANDS	4753
5	TAI PO	4911
6	WONG TAI SIN	5181
7	SAI KUNG	6258
8	NORTH	6699
9	TUEN MUN	6767
10	SHAM SHUI PO	7048
11	KWAI TSING	8236
12	SHA TIN	8765
13	YAU TSIM MONG	9045
14	EASTERN	10303
15	KOWLOON CITY	12581
16	TSUEN WAN	12624
17	YUEN LONG	15439
18	KWUN TONG	17515

() Set the location of a certain station in a district, such as MTR station, and find out the two nearest specific kinds of indoor sports facilities.

This is a practical question, for example, if we invite a friend to play badminton, but the two of us live far away, this question can help them find a badminton court in the middle district.

```

1  select address, fac_name, ST_Distance(ST_SetSRID(ST_MakePoint(114.1, 22.4), 4326),point(lat, lon)::geography) as distance
2  From badminton_courts
3  Where fac_type LIKE 'Sports Centres'
4  and district LIKE 'TAI PO'
5  order by ST_Distance(ST_SetSRID(ST_MakePoint(114, 22), 4326),
6  point(lat, lon))
7  limit 2;

```

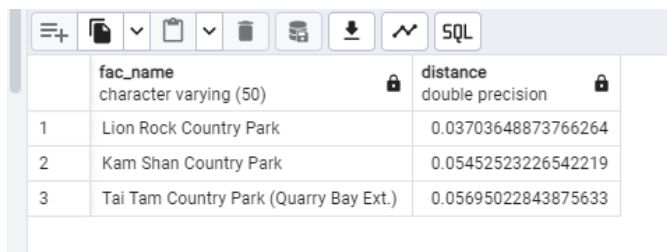
	address character varying (256)	fac_name character varying (256)	distance double precision
1	6/F, Tai Po Complex, 8 Heung Sze Wui Street, Tai Po	Tai Po Hui Sports Centre	8545.04324289
2	Podium Level, Tai Wo Shopping Centre, Tai Wo Estate, Tai ...	Tai Wo Sports Centre	8414.05820007

() Identify the 3 country parks that are closest to the geometric center (centroid) of the Kowloon City district.

Aim: to find the closest country parks to a given district using spatial queries.

First, we select the centroid of Kowloon City district from `hk_admin`, and the geometry points from country parks, then we perform a cartesian product after computing the distance, finally we order by the distance (ascending order) limit to 3 results. Since we are only interested in filtering to 3 country parks, it's acceptable for the distance to be arc degrees.

```
-- Identify the 3 country parks that are closest to the
-- geometric center (centroid) of the 'KOWLOON CITY' district
WITH kowloon_city_geom AS (
    SELECT ST_Centroid(geom) AS centroid
    FROM hk_admin
    WHERE district = 'KOWLOON CITY'
), country_parks AS (
    SELECT fac_name, point_hk1980(easting, northing) AS point
    FROM country_parks
)
SELECT cp.fac_name, ST_Distance(kcg.centroid, cp.point) AS distance
FROM country_parks cp
CROSS JOIN kowloon_city_geom kcg
ORDER BY distance
LIMIT 3;
```



	fac_name character varying (50)	distance double precision
1	Lion Rock Country Park	0.03703648873766264
2	Kam Shan Country Park	0.05452523226542219
3	Tai Tam Country Park (Quarry Bay Ext.)	0.05695022843875633

() Select all sports grounds that are top 2 in area in their district, and are greater than 25000 sq.m.

Aim: to find the sports grounds that are both big enough and stand out in their districts.

First, we select the district, facility name, area, and the ranking number of the sports ground area within the district, by using `over partition by` to separate the districts, and order by their areas in descending order. Finally, we filter the ranking number at most two and area > 25000.

```
-- Select all sports grounds 1st or 2nd largest in
-- their districts and areas greater than 25000 sqm
WITH district_sports_grounds AS (
    SELECT
        sg.district,
        sg.fac_name,
        sg.area,
        -- rank of sports ground area within district
        ROW_NUMBER() OVER (PARTITION BY sg.district ORDER BY sg.area DESC) AS rn
    FROM sports_grounds sg
)
SELECT dsg.district, dsg.fac_name, dsg.area
FROM district_sports_grounds dsg
WHERE dsg.rn <= 2 AND dsg.area > 25000;
```

	district character varying (30)	fac_name character varying (256)	area numeric (10,4)
1	EASTERN	Siu Sai Wan Sports Ground	43000.0000
2	KWAI TSING	Kwai Chung Sports Ground	43500.0000
3	KWUN TONG	Kowloon Bay Sports Ground	33320.0000
4	NORTH	North District Sports Ground	43000.0000
5	SAI KUNG	Tseung Kwan O Sports Ground	61738.0000
6	SHA TIN	Ma On Shan Sports Ground	41200.0000
7	SHA TIN	Sha Tin Sports Ground	29400.0000
8	SHAM SHUI PO	Sham Shui Po Sports Ground	44000.0000
9	SOUTHERN	Aberdeen Sports Ground	41800.0000
10	TAI PO	Tai Po Sports Ground	35790.0000
11	TSUEN WAN	Shing Mun Valley Sports Ground	34300.0000
12	WAN CHAI	Wan Chai Sports Ground	27000.0000
13	WONG TAI SIN	Hammer Hill Road Sports Ground	26000.0000
14	YUEN LONG	Tin Shui Wai Sports Ground	32200.0000

()

How many people aged >65 share one park or sport ground in highest population of age >65

```
--how many >65 share one park or sport ground in most population of age 65<
WITH age as (SELECT district FROM census
              order by age_5 DESC
              limit 1),
park as (
  (SELECT p.district FROM public.sports_grounds p,age a
   WHERE p.district = a.district)
  UNION ALL
  (SELECT p.district FROM public.park_zoos_and_gardens p,age a
   WHERE p.district = a.district)
),
num as (
  SELECT count(*) as count_num
  FROM park
)
SELECT a.district,(c.age_5::float/n.count_num) as average
FROM census c, num n, age a
WHERE c.district = a.district;
```

In this question, the target is to find out the highest population with age larger than 65 and how many of them may share a park or sports ground in that district.

In the SQL, age is used for finding out the most population with age larger than 65, and park is used to union sports_ground table and park_zoo_and_gardens table and limit to the district find in age. Then num is used to count the number of facilities in that district. District and the average

number of people share a facility is shown in the result.

	district [PK] character varying (30)	average double precision
1	KWUN TONG	29548.2

([REDACTED])

Find out the top 5 closest fitness center from the least number of fitness center district

```
WITH mid as (
  SELECT count(district) as num, district
  FROM public.fitness_rooms
  group by district
  ORDER BY num asc
  LIMIT 1
),
fitness as (
  SELECT *
  FROM public.fitness_rooms f
  WHERE f.district = (SELECT m.district FROM mid m)
  Limit 1
)
SELECT f.fac_name,pf.fac_name, st_distance (point(f.lat,f.lon)::geography,point(pf.lat,pf.lon)::geography) as distance
FROM fitness f,public.fitness_rooms pf
order by distance
limit 6;
```

The second question is to find out the top 5 closest fitness centers and the distance from the least number of fitness center district.

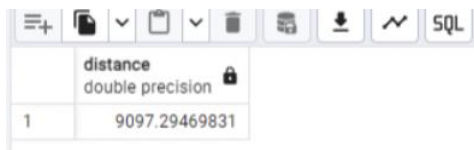
In the SQL mid is used to find out the least number of fitness center district and fitness is used to find out a fitness center in that district. Then we can find out the fitness center name and the distance and order them in ascending order.

	fac_name character varying (256)	fac_name character varying (256)	distance double precision
1	Lockhart Road Sports Centre	Lockhart Road Sports Centre	0
2	Lockhart Road Sports Centre	Hong Kong Park Sports Centre	1663.97305365
3	Lockhart Road Sports Centre	Kowloon Park Sports Centre	2690.15159652
4	Lockhart Road Sports Centre	Sheung Wan Sports Centre	2799.2551512
5	Lockhart Road Sports Centre	Kwun Chung Sports Centre	3010.84504515
6	Lockhart Road Sports Centre	Sun Yat Sen Memorial Park Sports Centre	3552.41767856

() My first SQL question is to find out how far the largest sports ground is from Tai Po District.

```
1 With I as (Select point(lat,lon)as point
2 From sports_grounds
3 order by area desc
4 limit 1),district as
5 (Select geom
6 from hk_admin
7 where district='TAI PO')
8 Select ST_distance(I.point::geography,district.
9 geom::geography)as distance
from I, district;
```

we can know the distance from Tai Po District to the largest sports ground in Hong Kong. The result is a distance of about 9097 meters.



	distance double precision
1	9097.29469831

() Another question is to find all swimming pool addresses in the most populated area.

```
1 with we as(select district,max(t_pop) as a
2 from census
3 group by district
4 order by a desc
5 limit 1)
6 select we.district, s.address, c.t_pop
7 from we,swimming_pools s, census c
8 where we.district=s.district and c.district=s.
9 district
```

since that area has the largest population, so overcrowding can be avoided while swimming. By finding the addresses of 3 swimming pools, let people know where the swimming pools are to spread out the crowd and let people have more fun.



	district character varying (30)	address character varying (256)	t_pop integer
1	SHA TIN	10 Yuen Wo Road, Sha Tin	692806
2	SHA TIN	33 On Chun Street, Ma On Shan	692806
3	SHA TIN	68 Che Kung Miu Road, Tai Wai	692806

() My first question is to find out the number of fitness room and the male ratio in different districts.

```
1 with i as (select district, count(district) as number_fitness_room
2 from fitness_rooms f
3 group by district)
4 select c.district, c.pop_m::float/c.t_pop::float *100 as percentage, i.number_fitness_room
5 from census c, i
6 where c.district=i.district
7 order by percentage desc
```

Highest Male Ratio are Tuen Mun (47.41%) and Yuen Long (47.01%) both have the highest male ratio with 5-6 fitness rooms each. Lowest Male Ratio is Wan Chai has the lowest male ratio (40.24%) with only 1 fitness room. Most Fitness Rooms is Kwai Tsing has the most fitness rooms (7) with a male ratio of 46.83%. Least Fitness Rooms is Wan Chai has the least fitness rooms (1).

Data Output Messages Notifications			
SQL			
	district [PK] character varying (30)	percentage double precision	number_fitness_room bigint
1	TUEN MUN	47.411906983718005	5
2	YUEN LONG	47.008741468087656	6
3	NORTH	46.9407133006708	5
4	KWAI TSING	46.83621152162776	7
5	KWUN TONG	46.39316305339248	5
6	TAI PO	46.14592220431636	5
7	WONG TAI SIN	46.098839238745136	3
8	SHAM SHUI PO	45.90178385024009	3
9	SHA TIN	45.88975846052142	5
10	ISLANDS	45.867920251292624	4
11	TSUEN WAN	45.7403137828263	4
12	SAI KUNG	45.61904313988512	4
13	YAU TSIM MONG	44.7044394441279	4
14	SOUTHERN	44.15066963437887	2
15	EASTERN	44.09755987031795	3
16	KOWLOON CITY	43.717276211906466	3
17	CENTRAL AND WESTERN	41.653210597025684	5
18	WAN CHAI	40.23815951288281	1

() My second question is to find out the distance between the sports grounds and park with zoos and gardens in the most populated district by age >65.

```
1 with p as(select district
2 from census
3 order by age_5 desc
4 limit 1 ),l as (
5 select *
6 from park_zoos_and_gardens z
7 where z.district = (SELECT district FROM p)), k as(
8 select *
9 from sports_grounds
10 where district=(SELECT district FROM p))
11 select l.fac_name,k.fac_name,st_distance (point(l.lat,l.lon)::geography,point(k.lat,k.lon)::geography) as distance
12 from p, l,k
13 order by distance
```

The most populated district for those aged over 65 is “Kwun Tong.” This district has only one sports ground, called “Kowloon Bay Sports Ground,” and four parks with zoos and gardens. “Ngau Tau Kok Park” is the closest park, at a distance of 695.74 meters from “Kowloon Bay Sports Ground,” while “Wai Lok Street Temporary Soccer Pitch” is the farthest park, at 3048.59 meters away.

Data Output Messages Notifications

	fac_name character varying (256)	fac_name character varying (256)	distance double precision
1	Ngau Tau Kok Park	Kowloon Bay Sports Ground	695.74117083
2	Jordan Valley Park	Kowloon Bay Sports Ground	1274.15937649
3	On Sau Road Park	Kowloon Bay Sports Ground	2331.28269033
4	Wai Lok Street Temporary Soccer Pitch	Kowloon Bay Sports Ground	3048.59356708